Division

In Python 2, division of integers defaults to integer division. In Python 3, it defaults to true division.

Python 2:

```
3/2   # 1
```

Python 3:

```
3/2   # 1.5
3//2  # 1
```

Print Statement

Print is a statement in Python 2 but a function in Python 3.

Python 2:

```
print "Hello"
```

Python 3:

```
print("Hello")
```

Unicode vs Strings

Python 2 strings are ASCII by default; Python 3 strings are Unicode.

Python 2:

```
u"hello"
```

Python 3:

```
"hello"  # already Unicode
```

Range / Xrange

Python 2 had both range (list) and xrange (iterator). Python 3 uses only range (iterator).

Python 2:

```
xrange(5)
```

Python 3:

```
range(5)
list(range(5))
```

Dictionary Methods

Dict methods return lists in Python 2 and views/iterators in Python 3.

Python 2:

```
mydict.keys()   # list
```

Python 3:

```
list(mydict.keys())
```

File Handling

Both versions use open(), but Python 3 requires explicit encoding for UTF-8.

Python 2:

```
f = open("doc.txt", "r")
raw = f.read()
```

Python 3:

```
f = open("doc.txt", "r", encoding="utf8")
```

```
raw = f.read()
```

String Formatting

Python 2 uses % formatting. Python 3 prefers format() or f-strings.

Python 2:

```
print "Value = %s" % var
```

Python 3:

```
print("Value = {}".format(var))
print(f"Value = {var}")
```

Iterators

map, zip, and range return lists in Python 2, but iterators in Python 3.

Python 2:

```
map(str, [1,2,3])   # ['1','2','3']
```

Python 3:

```
list(map(str, [1,2,3]))
```

NLTK Tokenizers

Code style is mostly unchanged between Python 2 and 3.

Python 2:

```
from nltk import word_tokenize
```

Python 3:

```
from nltk import word_tokenize  # same in Python 3
```